

Case Study

Black box design test cases techniques

1) equivalent class partition

Example 1 – Number Field

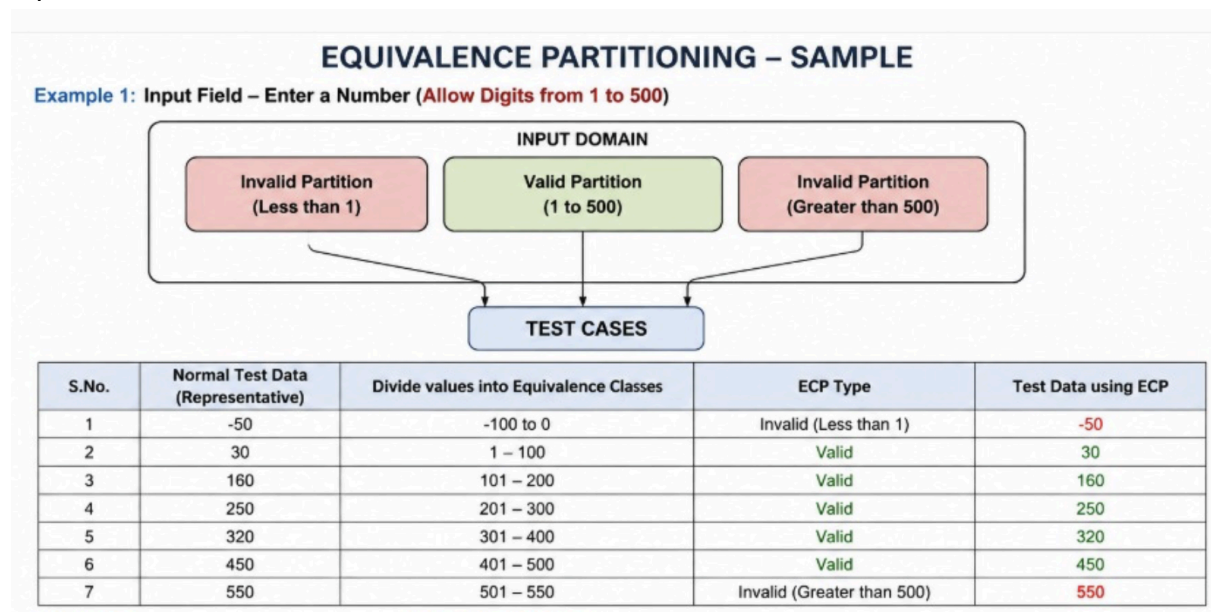
Design equivalence class partition for an application number field accepts 1-500 digits columns

1 serial no

2 normal test data

3 divide values into equivalent classes

Equivalence is divided into valid and invalid conditions



Example 2 – Name Field

The input field accepts **only alphabetic characters (A–Z, a–z)** for entering a name.

Numbers, special characters, alphanumeric values, and blank inputs are treated as invalid.

Example 2: Input Field – Enter a Name (Allow only alphabets)

S.No.	Normal Test Data (Representative)	Divide values into Equivalence Classes	ECP Type	Test Data using ECP
1	Anurag	Alphabets (A–Z, a–z)	Valid	Anurag
2	12345	Numbers (0–9)	Invalid	12345
3	@#\$\$%	Special Characters	Invalid	@#\$\$%
4	Anurag123	Alphanumeric (Alphabets + Numbers)	Invalid	Anurag123
5	(Blank)	Empty Input	Invalid	(Blank)

Example 3 – Email Field – Email ID Validation

This example uses equivalence partitioning to validate email addresses based on the presence of a **username**, **@ symbol**, and **domain**.

Missing components or empty inputs are grouped into invalid classes, while correctly formatted emails form the valid class

S.No.	Normal Test Data (Representative)	Divide values into Equivalence Classes	ECP Type	Test Data using ECP
1	userexample.com	Missing @ symbol	Invalid	userexample.com
2	user@.com	Missing domain name	Invalid	user@.com
3	user@domain	Missing top-level domain	Invalid	user@domain
4	user@domain.com	Valid email format	Valid	user@domain.com
5	user.name+tag@domain.co.in	Valid email with subdomain & tag	Valid	user.name+tag@domain.co.in
6	user @domain.com	Contains spaces	Invalid	user @domain.com
7	user@domain!.com	Invalid special character	Invalid	user@domain!.com

Example 4 – Password Field

The password must contain **at least 8 characters**, including uppercase, lowercase, a digit, and a special character.

Passwords that fail any of these requirements are classified as invalid inputs.

S.No.	Normal Test Data (Representative)	Divide values into Equivalence Classes	ECP Type	Test Data using ECP
1	abc12@	Less than 8 characters	Invalid	abc12@
2	abcdefg1@	Exactly 8 characters	Valid	abcdefg1@
3	abcd1234@#\$	Within 8 – 16 characters	Valid	abcd1234@#\$
4	abcdefghijklmnop	Exactly 16 characters	Valid	abcdefghijklmnop
5	abcdefghijklmnopq	Greater than 16 characters	Invalid	abcdefghijklmnopq
6	abc def12@	Contains space	Invalid	abc def12@
7	abc def	Contains invalid character	Invalid	abc def

Example 5 – Age Field – User Registration

This example demonstrates equivalence partitioning for an age input field where the valid range is **18 to 60 years**.

Inputs below 18 and above 60 are divided into separate invalid equivalence classes.

S.No.	Normal Test Data (Representative)	Divide values into Equivalence Classes	ECP Type	Test Data using ECP
1	-1	Less than 18	Invalid	-1
2	0	Less than 18	Invalid	0
3	17	Less than 18	Invalid	17
4	18	Lower boundary	Valid	18
5	35	Between 18 and 60	Valid	35
6	60	Upper boundary	Valid	60
7	61	Greater than 60	Invalid	61
8	100	Greater than 60	Invalid	100

Example 6 – Input Field – Select Country from Dropdown

This example demonstrates equivalence partitioning for selecting a country from a drop-down list. Valid inputs consist of countries present in the predefined list, while inputs not in the list, blank selections, or null values are classified as invalid.

S.No.	Normal Test Data (Representative)	Divide values into Equivalence Classes	ECP Type	Test Data using ECP
1	Atlantis	Not in dropdown list	Invalid	Atlantis
2	Narnia	Not in dropdown list	Invalid	Narnia
3	India	Present in list	Valid	India
4	United States	Present in list	Valid	United States
5	(Blank)	No selection made	Invalid	(Blank)
6	NULL	Null value	Invalid	NULL

Example 7 – Amount / Currency Field

The amount field accepts numeric values between **0.01 and 100,000.00** with up to 2 decimal places for an e-commerce transaction. Values less than 0.01 (including zero, negative numbers), values greater than 100,000.00, non-numeric characters, and values with more than two decimal places are considered invalid inputs.

S.No.	Normal Test Data (Representative)	Divide values into Equivalence Classes	ECP Type	Test Data using ECP
1	0	Less than 0.01	Invalid	0
2	-10.00	Less than 0.01	Invalid	-10.00
3	0.00	Less than 0.01	Invalid	0.00
4	0.01	Lower boundary	Valid	0.01
5	1234.56	Within range, 2 decimals	Valid	1234.56
6	100000.00	Upper boundary	Valid	100000.00
7	100000.01	Greater than 100000.00	Invalid	100000.01
8	123.456	More than 2 decimal places	Invalid	123.456
9	abc	Non-numeric value	Invalid	abc
10	12.3.4	Invalid numeric format	Invalid	12.3.4